KEMENTERIAN PENDIDIKAN

BAHAGIAN MATRIKULASI
MATRICULATION DIVISION

PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI
MATRICULATION PROGRAMME EXAMINATION

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RELATIVE ATOMIC MASSES OF SELECTED ELEMENTS
JISIM ATOM RELATIF UNSUR-UNSUR TERPILIH

Element	Symbol	Proton Number	Relative Atomic Mass
Aluminium	Al	13	27.0
Silver	Ag	47	107.9
Argon	Ar	18	40.0
Arsenic	As	33	74.9
Gold	Au	79	197.0
Barium	Ba	56	137.3
Beryllium	Be	4	9.0
Bismuth	Bi	83	209.0
Boron	B	5	10.8
Bromine	Br	35	79.9
Iron	Fe	26	55.9
Fluorine	F	9	19.0
Phosphorus	P	15	31.0
Helium	He	2	4.0
Mercury	Hg	80	200.6
Hydrogen	H	1	1.0
Iodine	I	53	126.9
Cadmium	Cd	48	112.4
Potassium	K	19	39.1
Calcium	Ca	20	40.1
Carbon	C	6	12.0
Chlorine	Cl	17	35.5
Cobalt	Co	27	58.9
Krypton	Kr	36	83.8
Chromium	Cr	24	52.0
Copper	Cu	29	63.6
Lithium	Li	3	6.9
Magnesium	Mg	12	24.3
Manganese	Mn	25	54.9
Sodium	Na	11	23.0
Neon	Ne	10	20.2
Nickel	Ni	28	58.7
Nitrogen	N	7	14.0
Oxygen	O	8	16.0
Platinum	Pt	78	195.1
Lead	Pb	82	207.2
Protactinium	Pa	91	231.0
Radium	Ra	88	226.0
Radon	Rn	86	222.0
Rubidium	Rb	37	85.5
Selenium	Se	34	79.0
Cerium	Ce	58	140.1
Cesium	Cs	55	132.9
Silicon	Si	14	28.1
Scandium	Sc	21	45.0
Tin	Sn	50	118.7
Antimony	Sb	51	121.8
Strontium	Sr	38	87.6
Sulphur	S	16	32.1
Uranium	U	92	238.0
Tungsten	W	74	183.9
Zinc	Zn	30	65.4

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Ionisation constant for water at 25 °C *Hasil darab ion bagi air pada* 25 °C	K_w	=	1.00×10^{-14} mol^2 dm^{-6}
Molar volume of gases *Isipadu molar gas*	V_m	= =	22.4 dm^3 mol^{-1} at STP 24 dm^3 mol^{-1} at room temperature
Speed of light in a vacuum *Laju cahaya dalam vakum*	c	=	3.0×10^8 m s^{-1}
Specific heat of water *Muatan haba tentu air*		= = =	4.18 kJ kg^{-1} K^{-1} 4.18 J g^{-1} K^{-1} 4.18 J g^{-1} °C^{-1}
Avogadro's number *Nombor Avogadro*	N_A	=	6.02×10^{23} mol^{-1}
Faraday constant *Pemalar Faraday*	F	=	9.65×10^4 C mol^{-1}
Planck's constant *Pemalar Planck*	h	=	6.6256×10^{-34} J s
Rydberg constant *Pemalar Rydberg*	R_H	= =	1.097×10^7 m^{-1} 2.18×10^{-18} J
Molar of gases constant *Pemalar gas molar*	R	= =	8.314 J K^{-1} mol^{-1} 0.08206 L atm mol^{-1} K^{-1}
Density of water *Ketumpatan air*	ρ	=	1 g cm^{-3}
Freezing point of water *Takat beku air*		=	0.00 °C
Vapour pressure of water *Tekanan wap air*	P_{water}	=	23.8 torr

UNIT AND CONVERSION FACTOR
UNIT DAN FAKTOR PERTUKARAN

Volume *Isipadu*	1 liter = 1 dm^3 1 mL = 1 cm^3
Energy *Tenaga*	1 J = 1 kg m^2 s^{-2} = 1 N m = 10^7 erg 1 calorie = 4.184 Joule 1 eV = 1.602×10^{-19} J
Pressure *Tekanan*	1 atm = 760 mm Hg = 760 torr = 101.325 kPa = 101325 N m^{-2}
Others *Lain-lain*	1 Faraday (F) = 96500 Coulomb (C) 1 Newton (N) = 1 kg m s^{-2}

Answer **all** questions.
*Jawab **semua** soalan.*

1 (a) Naturally occurring silver consists of two stable isotopes, ^{107}Ag and ^{109}Ag. The relative atomic mass of silver is 107.87. Show that isotope ^{107}Ag is more abundant.

Argentum yang wujud secara semulajadi mempunyai dua isotop stabil, ^{107}Ag *dan* ^{109}Ag. *Jisim atom relatif argentum ialah* 107.87. *Tunjukkan bahawa isotop* ^{107}Ag *adalah lebih banyak.*

[3 *marks*]
[3 *markah*]

(b) An amount of 500 g solution is prepared by dissolving 13.4 g of calcium chloride in water. Calculate the molarity and molality of the solution.

Sebanyak 500 g *larutan disediakan dengan melarutkan* 13.4 g *kalsium klorida dalam air. Hitung kemolaran dan kemolalan larutan tersebut.*

[4 *marks*]
[4 *markah*]

(c) Nitrate ion, NO_3^- can be reduced to nitrous oxide, N_2O by anaerobic bacteria in a redox process. During this process, water is also oxidised producing oxygen. Write a balanced equation for this reaction under alkaline condition.

Ion nitrat, NO_3^- *boleh diturunkan kepada nitrus oksida,* N_2O *oleh bakteria anaerobik dalam proses redoks. Semasa proses ini, air turut teroksida menghasilkan oksigen. Tulis persamaan berimbang untuk tindak balas ini dalam keadaan beralkali.*

[5 *marks*]
[5 *markah*]

(d) Aluminium chloride, $AlCl_3$ is produced from aluminium metal in the presence of hydrogen chloride, HCl. The equation for the reaction is as follows:

Aluminium klorida, $AlCl_3$ *dihasilkan dari logam aluminium dalam kehadiran hidrogen klorida,* HCl. *Persamaan bagi tindak balas tersebut adalah seperti berikut:*

$$2Al + 6HCl \rightarrow 2AlCl_3 + 3H_2$$

(i) Determine the amount of $AlCl_3$ produced (in g) if 2.67 g Al reacts with 3.5 g HCl.

Tentukan jumlah $AlCl_3$ *yang dihasilkan* (*dalam* g) *jika* 2.67 g Al *bertindak balas dengan* 3.5 g HCl.

(ii) Calculate the percentage yield for this reaction if 3.50 g of $AlCl_3$ is produced.

Hitung peratusan hasil bagi tindak balas ini jika 3.50 g $AlCl_3$ *terhasil.*

[9 *marks*]
[9 *markah*]

2 (a) The electron transition in the Balmer series of hydrogen atom from n=4 to n=2 is shown as a line spectrum.

Peralihan elektron dalam siri Balmer bagi atom hidrogen dari n=4 ke n=2 adalah ditunjukkan sebagai spektrum garisan.

(i) Calculate the energy associated with this transition.

Hitung tenaga yang berkaitan dengan peralihan ini.

(ii) Explain whether energy is emitted or absorbed during the transition.

Jelaskan sama ada tenaga dibebaskan atau diserap semasa peralihan tersebut.

(iii) Determine the wavelength for this transition.

Tentukan panjang gelombang bagi peralihan ini.

[6 *marks*]
[6 *markah*]

(b) Chromium and manganese have the same number of electrons in the 3*d* orbital. Explain this statement in relation to the filling of orbitals with electrons.

Kromium dan mangan mempunyai bilangan elektron yang sama dalam orbital 3d. Terangkan penyataan ini dengan menghubungkan pengisian orbital dengan elektron.

[4 *marks*]
[4 *markah*]

3 (a) The molecules HCN, CH_3NH_2 and CH_2NH each contains nitrogen and show different chemical behaviours.

Molekul HCN, CH_3NH_2 *dan* CH_2NH *masing-masing mengandungi nitrogen dan menunjukkan sifat kimia yang berbeza.*

(i) Draw the Lewis structure for each molecule.

Lukis struktur Lewis bagi setiap molekul.

(ii) Arrange the molecules in order of increasing bond length of C–N.

Susun molekul mengikut tertib panjang ikatan C–N *menaik.*

[4 *marks*]
[4 *markah*]

(b) Compounds ICl_3 and PCl_5 can exist in the gaseous state, however they exhibit different behaviour.

Sebatian ICl_3 *dan* PCl_5 *boleh wujud dalam keadaan gas, walau bagaimanapun mereka mempamerkan sifat berbeza.*

(i) Predict the molecular shapes and bond angles in each compound.

Ramalkan bentuk molekul dan sudut ikatan dalam setiap sebatian.

(ii) State the hybridisation of the central atom for each compound.

Nyatakan penghibridan atom pusat bagi setiap sebatian.

[8 *marks*]
[8 *markah*]

(c) Both calcium and potassium are metals. Describe the bonding in these metals and compare their strength.

Kalsium dan kalium kedua-duanya adalah logam. Huraikan ikatan dalam logam-logam tersebut dan bandingkan kekuatannya.

[5 *marks*]
[5 *markah*]

4 (a) The density of a gaseous fluoride of phosphorus is 3.91 g L^{-1} at STP. Calculate the molar mass of this fluoride and determine its molecular formula.

Ketumpatan gas fluorida bagi fosforus ialah 3.91 g L^{-1} *pada* STP. *Hitung jisim molar fluorida ini dan tentukan formula molekulnya.*

[5 *marks*]
[5 *markah*]

(b) **TABLE 4** shows the vapour pressures of several alcohols at 20 °C.

JADUAL 4 *menunjukkan tekanan wap beberapa alkohol pada* 20 °C.

TABLE 4
JADUAL 4

Compound ***Sebatian***	**Vapour pressure / atm** ***Tekanan wap* / atm**
CH_3OH	0.117
C_2H_5OH	0.059
C_3H_7OH	0.026
C_4H_9OH	0.006

Based on the data in **TABLE 4**, arrange the alcohols in the order of increasing boiling points. Explain.

Berdasarkan data dalam ***JADUAL 4****, susun alkohol tersebut dalam tertib takat didih menaik. Jelaskan.*

[4 *marks*]
[4 *markah*]

5 (a) A flask containing 0.60 M of Br_2 and 0.60 M of Cl_2 are allowed to reach equilibrium. After equilibrium is achieved, the flask is found to contain 0.28 M of BrCl. Calculate the value of K_c for this reaction.

Satu kelalang mengandungi 0.60 M Br_2 *dan* 0.60 M Cl_2 *dibiarkan mencapai keseimbangan. Selepas keseimbangan dicapai, kelalang didapati mengandungi* 0.28 M BrCl. *Hitung nilai* K_c *bagi tindak balas ini.*

[5 *marks*]
[5 *markah*]

(b) Water splitting is a pathway to produce clean hydrogen fuel. The reaction is given below.

Pembelahan air adalah laluan untuk menghasilkan bahan api hidrogen bersih. Tindak balas tersebut diberikan di bawah.

$$2H_2O(l) \rightleftharpoons 2H_2(g) + O_2(g) \qquad \Delta H = +285.8 \text{ kJ mol}^{-1}$$

Predict what will happen to the system and equilibrium constant if the temperature is increased.

Ramalkan apa yang akan berlaku kepada sistem dan pemalar keseimbangan jika suhu ditingkatkan.

[4 *marks*]
[4 *markah*]

6 (a) Propanoic acid, C_2H_5COOH, is used in the manufacture of calcium propanoate, a food preservative. Calculate the pH of the solution and the degree of dissociation if the initial concentration of C_2H_5COOH is 0.658 M. [Acid dissociation constant, K_a for C_2H_5COOH is 1.34×10^{-5}]

Asid propanoik, C_2H_5COOH, *digunakan dalam pembuatan kalsium propanoat, sejenis pengawet makanan. Hitung* pH *larutan dan darjah penceraian jika kepekatan awal* C_2H_5COOH *ialah* 0.658 M. [*Pemalar penceraian acid*, K_a *bagi* C_2H_5COOH *ialah* 1.34×10^{-5}]

[6 *marks*]
[6 *markah*]

(b) Determine whether a precipitate will form if 0.5 L solution of 0.012 M $Fe(NO_3)_2$ is mixed with 0.8 L solution of 0.002 M Na_2S. [Solubility constant, K_{sp} for FeS is 4.9×10^{-18}]

Tentukan sama ada mendakan akan terbentuk jika 0.5 L *larutan* 0.012 M $Fe(NO_3)_2$ *dicampur dengan* 0.8 L *larutan* 0.002 M Na_2S. [*Pemalar keterlarutan*, K_{sp} *bagi* FeS *ialah* 4.9×10^{-18}]

[8 *marks*]
[8 *markah*]

END OF QUESTION PAPER
KERTAS SOALAN TAMAT